2 (a) $E = hc / \lambda = (6.63 \times 10^{-34} \times 3.0 \times 10^{8}) / (486 \times 10^{-9})$ C1

$= 4.09 \times 10^{-19}$ J ...(allow 2 sf).. A1 [2]

(b) energy level drawn at 4.09×10^{-19} J B1

transition 4.09×10^{-19} to zero clear B1

transition 4.09×10^{-19} to 3.03×10^{-19} clear .... B1 [3]

(-1 for reversed arrows, -1 for extra level at 1.06)

3 (a) (i) constant amplitude B1